

Nagarjunan Saravanan

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EDUCATION

Binghamton University, SUNY, Thomas J. Watson College of Engineering and Applied Science, Binghamton, NY

Master of Science in Computer Science, Artificial Intelligence Track (GPA: 3.71/4.0)

Aug 2024 – May 2026

Relevant Coursework: Artificial Intelligence, Machine Learning, Computer Vision, Robot Perception.

Madras Institute of Technology, Anna University, Chennai, TN, India

Bachelor of Engineering in Electronics and Communication

Aug 2014 – May 2018

Relevant Coursework: Data Structures, Operating Systems, Object Oriented Programming

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, ESQL, C++, C, JavaScript, HTML, CSS

Machine Learning & AI: PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, DistilBERT, BERTweet, XGBoost, LightGBM

Deep Learning & Computer Vision: CNN/RNN/LSTM, YOLOv8, G-DINO, OWL-ViT, OpenCV, Obj Detection, Pose Estimation, OpenNSFW2.

Data Science & Analytics: NumPy, Pandas, Matplotlib, Seaborn, Plotly, Jupyter, Time Series, A/B Testing, Data Visualization

Cloud & Deployment: AWS (CCP), Google Cloud Run, Docker, FastAPI, REST APIs, CI/CD (GitHub Actions), Linux

Databases & Tools: MySQL, Power BI, MS Excel, Git/GitHub, SOAPUI, IBM v10, IBM ACE, IBM MQ, Putty, WinSCP, Jenkins

PROFESSIONAL EXPERIENCE

Uplifty AI, AI & Machine Learning Algorithm Engineering Intern | Austin, Texas (Remote)

Aug 2025 – Present

Tech Stack: PyTorch, Transformers, BERTweet, DistilBERT, LoRA, Scikit-Learn, OpenAI API, FastAPI, Supabase, Google Cloud Run, NLP.

- Led the **Human Growth Index (HGI)** team, building a **multi-model NLP** pipeline (**BERTweet + LoRA + mixed-precision**) using a **100k+** reflection dataset, boosting accuracy to **95%** and enabling scalable **0–100** wellness scoring across **13** dimensions.
- Engineered Uplifty's production AI moderation system by integrating rules, **TF-IDF** baselines, and a fine-tuned **DistilBERT-RNN** model, delivering real-time moderation on Cloud Run with **96%+** auto-flagging and full **Supabase** workflow integration.

State Street Corporation, Senior Associate | Bangalore, KA, India

Jul 2022 – Aug 2024

Tech Stack: Python, Numpy, Pandas, Matplotlib, Seaborn, MS Excel, Power BI, IBM ACE, IIB v10, ESQL, MQ, Putty, WinSCP

- Built and optimized **EMEA** integrations using **IBM ACE**, improving reporting accuracy by **30%** and analyzing **10,000+ SWIFT** transactions while delivering UAT/production reports and **6+ Power BI** dashboards for insight-driven decisions.
- Transformed **OTC** derivatives data from **100+** client sources (**400+ products**) and automated **80%** of routine data prep with Python, reducing manual workload by **~25 hours/month** and strengthening global financial workflows.

Tata Consultancy Services, System Engineer | Chennai, TN, India

Sep 2018 – Jun 2022

Tech Stack: IBM ACE, IIB v10, ESQL, SOAPUI

- Integrated **Guidewire PolicyCenter** with **OpenText Exstream** via **IIB v10**, delivering **200–300** forms per Commercial LOB across two production releases and enabling seamless interactions with **Claims, Rating, and Comparator** services.
- Designed a document-production **POC** that reduced processing time from **200ms** to **4ms**, while managing **ESQL** transformations, **SOAPUI** testing, and **Agile** delivery improvements across enterprise workflows.

RESEARCH EXPERIENCE

Research Assistant, Marine Robotics Lab | Binghamton University

Aug 2025 – Present

Tech Stack: BlueBoat USV, ROS Noetic, Sonar (SSS/SBE), Sensor Fusion, Python, OpenCV, Linux, Docker, OpenCV, BlueOS.

- Conducted field deployments using **BlueBoat USV** to collect sonar data from **side-scan** and **single-beam echosounders**, supporting **bathymetric mapping** and **marine perception research** under the **Marine Robotics Lab**.
- Installed and configured **ROS Noetic** for **sensor-fusion** development; currently researching and prototyping **data-fusion algorithms** integrating sonar modalities to enhance underwater mapping accuracy.

PROJECTS

Open-Vocabulary Object Detection for Maritime Perception, CV & Robotics | Binghamton University

Sep 2025 – Nov 2025

Tech Stack: PyTorch, Transformers, YOLOv8, Grounding DINO, OWL-ViT, OpenCV, NumPy, Matplotlib.

- Architected** a hybrid open-vocabulary detection pipeline using **YOLOv8**, **OWL-ViT**, and **Grounding DINO**, enabling **zero-shot recognition** of unseen maritime obstacles across MaSTr1325 and MODD datasets.
- Achieved **mAP 0.71 (+24 % recall)** and demonstrated **language-driven visual grounding** (“detect kayaks,” “detect debris”), advancing **foundation-model perception** for Autonomous surface vessels.

Augmented Reality with Planar Homographies, Computer Vision | Binghamton University

Sep 2025

Tech Stack: Python, OpenCV, NumPy, SciPy, scikit-image, Homography, RANSAC, FAST/BRIEF, Image Warping & Video Processing.

- Constructed** a complete AR pipeline integrating **FAST/BRIEF**, **20-iteration RANSAC homography**, and descriptor caching matching **150–350 features/frame** and delivering **~3× faster** real-time tracking over **300+ frames**.
- Engineered robust **AR video overlays** with **90% jitter reduction** using **homography smoothing ($\alpha=0.9$)**, **perspective warping**, and **frame-adaptive cropping**; also directed seamless **2×-width panoramas** via **automatic feature stitching**.

Demand Forecasting System for Retail, ML, Time-Series, Deep Learning | Independent Project

May 2025 - Sep 2025

Tech Stack: Python, Scikit-Learn, XGBoost, LightGBM, Prophet, PyTorch (TFT), FastAPI, Streamlit, Docker, Time-Series.

- Designed** a full forecasting pipeline with **40+ features** across **125K+ records**, benchmarking LightGBM, XGBoost, Prophet, and TFT; where TFT delivered the best validation (RMSE **443.7**) and Prophet the most interpretable baseline (RMSE **211.0**).
- Deployed a **FastAPI + Docker + Streamlit** pipeline delivering **<60ms** predictions and **95%** faster cached **TFT** inference, with unified **RMSE/MAPE** backtesting and scalable **multi-model JSON forecasting**.